



Geothermal
Technical Section



Society of Petroleum Engineers

Geothermal Technical Section

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NEWSLETTER

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FROM THE SECTION CHAIR

Dear fellow GTTS members,

I am thrilled to update you about the “State of the Union” of the section.

First, I am very thankful that the GTTS Board has an excellent group of volunteers this year who are enthusiastic about geothermal. They all have been actively and proactively participating and contributing. I can vouch that they are NOT serving for the money, power, or fame. They are an exceptional group of professionals who truly serve for the best interest of the Technical Section and the Geothermal community. For their outstanding teamwork and the organization culture, I am very thankful and exceedingly honored to serve with them.

Below are seven key GTTS achievements this year.

- (a) **Submittal of Annual Report:** The successful submittal of our annual report in time qualifies us to be considered by the SPEI award judges for a Technical Section Award. I thank the contributions by several board members, especially Gabrijel Grubac who coordinated in information gathering and drafted the first version. I hope we will win an award this year. Please wish us luck!
- (b) **Membership Growth:** GTTS membership has grown tremendously during last 12 months. Professional membership grew by ~130% (1,959 vs 4,550) and our student membership grew by a staggering ~300% (from 324 vs 1,316) as of April 29, 2026. Kudos to Kathleen Gould and the GTTS Regional Membership Chairs and other board members.
- (c) **Monthly Webinars:** As a key benefit to our members, we offer at least one free online seminar monthly when we invite experts to present geothermal technical information. The format can be solo presentation, panel discussion or co-hosting with other Technical Sections. It takes a lot of work to identify and invite speakers, coordinate, announce and host the presentation. Thanks to the demanding work by our Program Chairs: Katerina Yared (2026) and Gabrijel Grubac (2025).
- (d) **Quarterly Newsletters:** Another key benefit to our members is our award-deserving newsletter. It is highly informative and shares up to 38 pages of geothermal news and articles. I am immensely proud of the quality of these publications and appreciate the incredible work of our newsletter team: Sonia Isabella Lopez Kovacs, Vinod Kumar Madem, and Mahmoud Algaier.

FROM THE SECTION CHAIR

- e) **Geothermal Event Sponsorships:** The Board initiated an effort to recruit members at three key geothermal events: SPE Europe Workshop in Milan, Italy, the premier World Geothermal Congress at Calgary, Canada and the Africa Geothermal Workshop at Nairobi, Kenya. Thanks to Devindra Harshad Rai, GTTS Treasurer, who worked hard in managing the budget and volunteered to lead the booth booking and recruiting effort at the Calgary event. I also thank Mahesh Jayaraman and Jill Thomas, SPEI staff based in Calgary who helped us welcome conference attendees and told them about GTTS. Thanks to Princewill Okechukwu for stepping in and giving a talk at the Milan event. Thanks in advance to Reza Khastoo for helping us to host a networking evening at the Kenya event.
- f) **New Webpage:** Our new webmaster, Amr Ramadan, has totally revamped our webpage. He has added many features, please go and check it out.
- g) **Collaboration with other SPE Technical Sections:** GTTS Technical Directors have started an effort to explore collaborations with other SPE Technical Sections. They are planning on co-hosting webinars.

There are four items coming to you soon. (a) Student Scholarships. (b) GTTS Awards (c) Networking at ATCE, and (d) Online Geothermal Training Class. Please keep an eye out for these announcements.

In announcing the 2026-2027 Student scholarships, I need to specially thank the sponsors of our 2025 ATCE Networking Event whose support has allowed us to offer this year's student scholarships. Many thanks to XGS Energy, QuantumPro, Geolog, SLB, Carbo, and Blade Energy Partners. On behalf of the students, we thank you!

Fellow members, we will need additional volunteers next year! If you are interested to serve on the Board or committees, please inform me at steveior@yahoo.com.

Let us work together to make GTTS even better, to expand the geothermal eco-system and to transfer geothermal related knowledge.

Steve Cheung Ph.D.
Chairperson, SPE Geothermal Technical Section



SPE GEOTHERMAL TECHNICAL SECTION

The **SPE Geothermal Technical Section** (SPE GTTS) is a forum for SPE members who share a common interest in geothermal energy and its potential to transform electric power generation and direct use of applications around the world.

The SPE GTTS also focus on the transfer of technology and practices from the oil and gas industry to geothermal energy development and promote industry best practices.

Geothermal Technical Section



SPE GTTS seeks to:

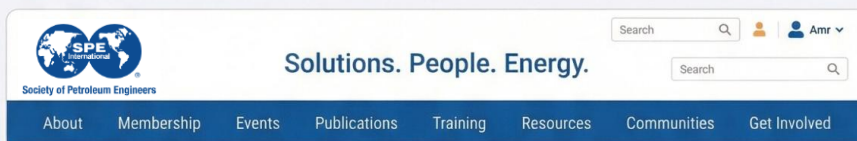
- Collect, disseminate, and exchange technical knowledge concerning geothermal energy, and demonstrate its proximity to more traditional oil and gas disciplines.
- Introduce opportunities to grow careers and businesses by knowledge and skills transfer.
- Establish collaboration across industry and promote industry best practices
- Society of Petroleum Engineers (SPE) and SPE GTTS, are actively engaging through knowledge and technology sharing and transfer to support alternatives and technologies, to a sustainable and low carbon energy transition in the oil and gas industry.



<https://connect.spe.org/gtts/home>

How to Register for the Geothermal (GTTS) Technical Section

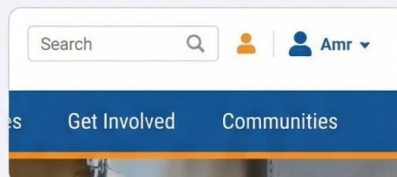
1. Go to spe.org



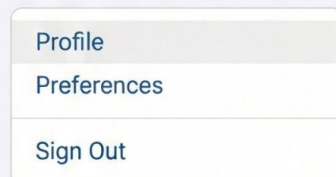
Supporting Detail: Visit the official Society of Petroleum Engineers homepage to begin the registration process.

2. Login

Supporting Detail:
Access your personal member account by clicking the login icon or your name in the top right corner of the page.



3. Click on the 'Profile' option



Supporting Detail:
Open the account dropdown menu and select "Profile" to manage your membership details and preferences.

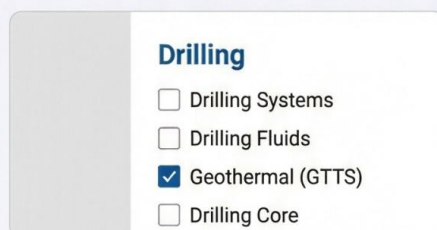
4. Scroll down to 'Technical Sections'

Supporting Detail:
Locate the "Preferences" menu on the left-hand sidebar and select the "Technical Sections" link.

Preferences

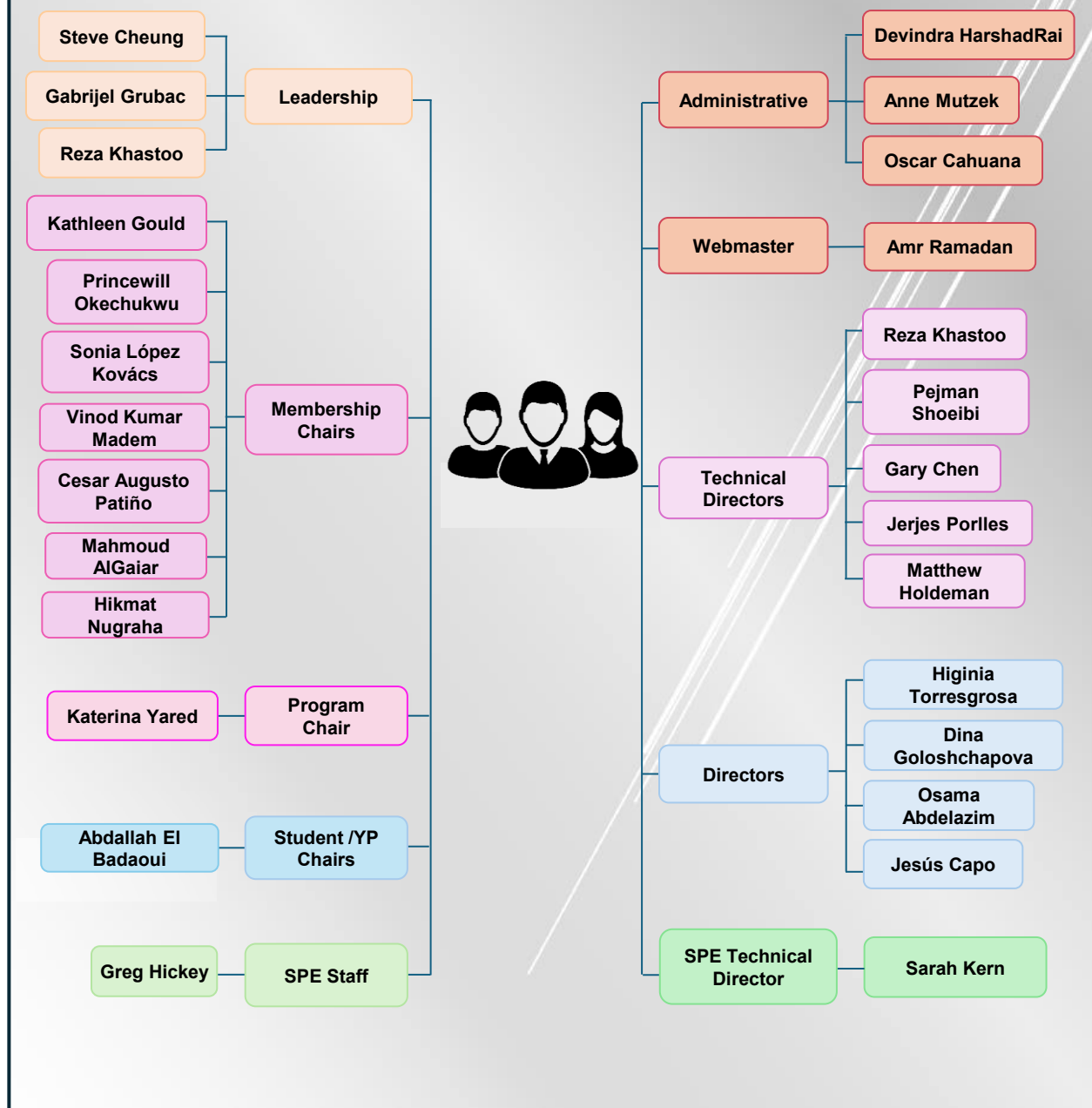
Communication Preferences
Technical Disciplines
Technical Sections
Volunteering Interest
Awards & Recognition
Mentoring

5. Find and select 'Geothermal (GTTS)'



Supporting Detail:
Within the list of technical categories, find the "Drilling" section and check the box for "Geothermal (GTTS)".

SPE GEOTHERMAL TECHNICAL SECTION OUR TEAM



Call for Nominations – Recognizing Excellence in Geothermal Energy

The SPE Geothermal Technical Section (GTTS) is pleased to launch its **annual Awards Nomination Campaign**. This initiative aims to recognize outstanding professionals who have made significant contributions to geothermal energy, technical innovation, industry leadership, and service to the SPE community. Our goal is to build a strong pipeline of high-quality nominations and increase geothermal representation among SPE award recipients

We invite GTTS members and beyond to identify and nominate deserving colleagues whose work has advanced our industry and inspired others through technical excellence, innovation, mentorship, or volunteer leadership.

Nomination Period: 1 July - 30 August



During this period, GTTS will collect candidate recommendations for relevant SPE awards and recognition programs. Nominations may **include professionals from industry, academia, research organizations**, and the broader geothermal community.

Why Nominate? Awards provide an opportunity to:



Recognize **exceptional technical and professional achievements**. Highlight the **growing impact of geothermal energy** within the global energy transition. **Celebrate individuals** who contribute to the advancement of our industry and the SPE community. Increase visibility for geothermal professionals at the international level.

Review Process



All nominations will undergo an **independent review process** conducted by a dedicated GTTS review panel. To ensure fairness and objectivity, nominations will be evaluated using a standardized scoring matrix based on technical impact, industry contribution, professional service, leadership, and innovation.

We encourage all members to participate by:

- Submitting candidate recommendations.
- Promoting the nomination campaign within their professional networks.
- Helping identify emerging and established leaders in the geothermal sector.

Additional information on nomination criteria, deadlines, and submission procedures will be shared through GTTS communication channels and LinkedIn throughout the nomination period.

Let us work together to recognize the individuals who are shaping the future of geothermal energy.

Submit your nomination today!



Scan the QR code or click [here](#) to recommend outstanding professionals from the geothermal community.

Nomination deadline: **30 August**.

Sponsorship Opportunities

The **SPE Geothermal Technical Section (GTTS)** invites you to partner with us as a 2026/2027 sponsor.

This is an opportunity to position your organization within a focused technical community advancing geothermal energy, innovation, and cross-industry collaboration. GTTS brings together professionals, operators, service companies, technology providers, researchers, and energy leaders who are shaping the future of geothermal development.

Your sponsorship would support knowledge sharing while giving your organization meaningful visibility across our in-person and digital engagement channels. In addition, it would allow GTTS to continue supporting students through the scholarship program as done in the past years.

WHY PARTNER WITH GTTS?

- **Brand Alignment**
→ Associate your company with geothermal technical leadership.
- **Community Access**
→ Engage professionals across events, webinars, publications, and digital channels.
→ Become part of the supportive pillars for the continuation of geothermal education and momentum across the energy world.

SPONSORSHIP Packages:

- There are Tier 1 (\$4000) and Tier 2 (\$2000) packages, which feature ATCE benefits plus GTTS complimentary benefits. See next page for details!

Sponsorship Contract Deadline is 4th September. For Sponsorship Details and Funds, see more details on:

<https://atce2026.smallworldlabs.com/exhibitor/sponsorship-items>

For More Sponsorship Information, connect with us through:

Steve Cheung; Osama Abdelziem

Sponsorship Opportunities

Sponsorship Tier	Tier 1	Tier 2
Sponsorship Package	\$4,000	\$2,000
ATCE 2026 Package		
🏠 On-Site Advantages		
ATCE 2026 GTTS Reception Tickets	4 Tickets	2 Tickets
High-Visibility On-Site Brand Positioning (ATCE 2026 Signage)	Included	Included
Promotional Showcase at the Networking Reception	Included	Included
⊕ Online Advantages		
Logo on ATCE 2026 Website	Included	-
Recognition & Logo on SPE Connect GTTS	Included	-
Dedicated Logo Spot in Technical Section Email Campaign	Included	Included
Complimentary Package		
🔍 Interview Feature		
5-Minute Interview during ATCE 2026	Included	-
💻 Virtual Events & Webinars		
Host a Company-Related Webinar	1 / Year	-
Recognition of Sponsor on Monthly Webinars	3 Webinars	2 Webinars
📰 Digital Publications		
Articles in GTTS Quarterly Newsletter	1 / Year	-
Recognition in Newsletter	Included	Included
📱 Mobile App Engagement		
Company Promotional Materials on GTTS App Push Notification	2 / Year	1 / Year
Logo on GTTS Mobile App	Included	Included
📌 Social Media Outreach		
Dedicated LinkedIn Posts	2 / Year	1 / Year
Joint Sponsor Recognition Post	1 / Year	1 / Year

For More Sponsorship Information, connect with us through:
[Steve Cheung](#); [Osama Abdelziem](#)

Geothermal Well Control: Complications Beyond Conventional Oil and Gas Practices

*by Guizhong Chen, NOV; Toney Deer, Well Control School;
Chuck Wright, Jessica Stump, NOV*

Geothermal well control has unique challenges compared with conventional oil and gas (O&G) because of high temperatures and fluid phase changes. Geothermal well control requires adaptation beyond conventional O&G practices in well construction, well integrity, high-pressure/high-temperature (HP/HT) equipment selection, corrosive environments, wellbore fluid phase change behavior, kick identification and shut-in methods, and methods to regain control of the well.

Conventional O&G well control relies on maintaining dynamic hydrostatic pressure above the formation pressure. In O&G, gas has predictable behavior, while in geothermal systems, flashing occurs when wellbore pressure drops below the water saturation pressure at a given temperature (Fig. 1), reducing fluid density (Fig. 2), and potentially leading to an aquathermal event (steam blowout) without a formation influx. The common solution for flashing or aquathermal events is quenching the well by circulating cold water into the wellbore. However, this can result in the steam column collapsing too quickly and potentially creating a sudden relative vacuum in the wellbore.

Flashing can trigger an abrupt and violent steam expansion on the rig floor during drilling or other well operations (Deer, 2025). An expansion of ~1,600 times in volume at the atmosphere condition can be calculated as the ratio of the specific volume of saturated vapor to saturated liquid at 100°C and 1 atm (Keenan et al., 1978, page 3; Lardner, 1840, page 117; Ripley and Dana, 1879). When water changes from a liquid to a gaseous state (steam), a cubic inch of water evaporated under the ordinary atmospheric pressure is converted into nearly a cubic foot of steam.

Drilling conventional hydrothermal wells typically involves highly permeable and highly fractured formations (e.g., conventional hydrothermal, volcanic, and/or sedimentary), often leading to lost returns during drilling (Umam et al. 2020). Losing circulation can be an indicator of a potential feed zone for the well (Winn et al. 2023). Severe lost circulation commonly indicates intersection of highly permeable fracture zones, which usually correspond to productive reservoir feed zones, although confirmation through temperature and sustained flow analysis is required. Fluid loss management involves securing and managing the availability of a sufficient and reliable water supply and/or a high volume of high-temperature drilling fluids. A sudden fluid level drop due to severe lost returns during drilling will quickly lower the hydrostatic and bottomhole pressures, creating a high potential for flashing/thermal kicks.

The positive indicators from O&G kick detection are still valid in geothermal operations if a full column of fluid is available. However, drilling blind is common in geothermal. Regardless of whether lost circulation is present, distinguishing common thermal kick indicators becomes critical. Common indicators include but are not limited to rapid increase in return fluid temperature, steam showing at the surface, audible boiling sound coming from the annulus, and wellbore flashing without pore fluid in the reservoir. Due to the speed of wellbore fluid phase change behaviors, shut-in methods may need to be adjusted from conventional practice so that a flashing event or thermal kick can be diverted. Quenching the wellbore may be required to regain control of the well and return it to a thermally stable state.

Geothermal Well Control: Complications Beyond Conventional Oil and Gas Practices (Cont.)

*by Guizhong Chen, NOV; Toney Deer, Well Control School;
Chuck Wright, Jessica Stump, NOV*

Developing the geothermal well control guidelines and training curriculum is essential to today's fast growing geothermal drilling industry, especially for the hot dry rock (~150–350°C, Anderson et al. 2025), superhot rock (>374°C, Hill et al. 2025) and supercritical rocks (>374°C and >22 MPa, CATF). The training material needs to be tailored to not only geothermal well drilling but also the entire well life cycle. As shown in Fig. 3, using Boyle's gas law to simulate steam behavior can create erroneous results. Water vapor can be treated as an ideal gas at pressures below 10 kPa; however, at high pressures the ideal gas assumption yields unacceptable errors, particularly in the vicinity of the critical point and the saturated vapor line (over 100 percent) (Cengel et al. page 127, Fig. 4-45). A theoretically sound, well-calibrated, industry-recognized well control simulator and a geothermal-specific well control certification and re-certification course are strongly advised to be required for the fast growing high-temperature, superhot and supercritical geothermal drilling activities. Learning from past well control incidents is also critical to benefit from real-world lessons (Ranka and Deer, 2025).

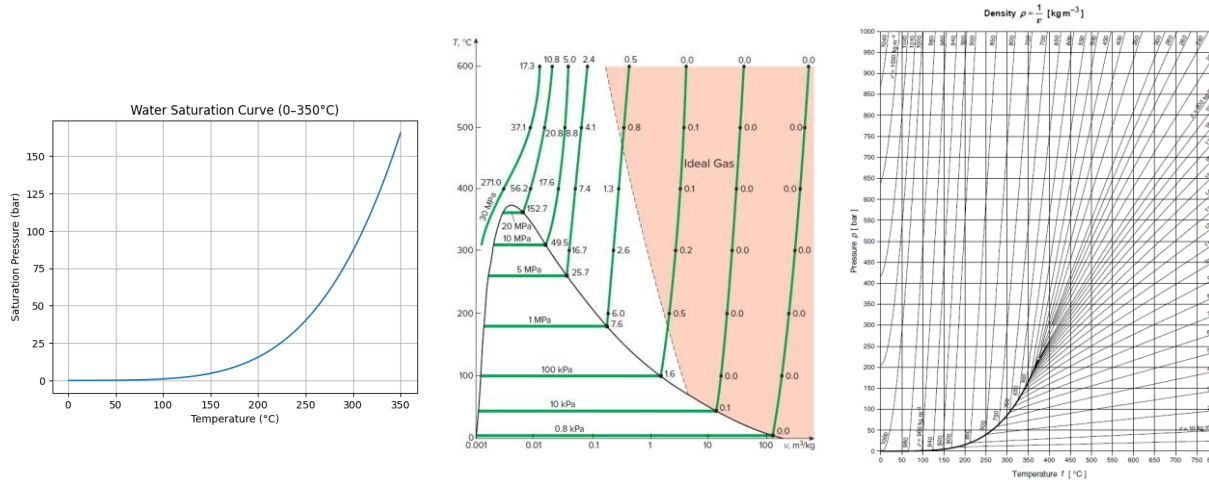


Fig. 1 (left) Water saturation curve (Ref: **Engineering toolbox**). Below the saturation curve, the fluid is in a steam state while above the curve the fluid is in a liquid state.

Fig. 2 (middle) Pressure-temperature diagram with lines of constant density.

Fig. 3 (right) Percentage of error involved in assuming steam to be an ideal gas (**Cengel et al. page 127, Fig. 4-45**)

Geothermal Well Control: Complications Beyond Conventional Oil and Gas Practices (Cont.)

*by Guizhong Chen, NOV; Toney Deer, Well Control School;
Chuck Wright, Jessica Stump, NOV*

Geothermal well control often demands high-temperature equipment (e.g., BOP and wellhead) with 300°C+ rated elastomer seals, managed pressure drilling (MPD) or diverter equipment for steam management, and BOP stack components for steam diverting and steam shut in. Casing for geothermal wells needs to incorporate all the thermal loads and operational cyclic loads, in addition to the O&G well casing design requirements. The H₂S and CO₂ gases are often present in geothermal systems, and the H₂S concentration increases approximately five-fold for every 50°C increase in temperature (Blamey, 2006). Therefore, the identification of corrosive H₂S and CO₂ gases is critical when detecting and interpreting gas kicks (non-condensable gases) in a geothermal well.

The rapidly increasing energy demand, low-carbon energy market, and fast-paced government funding around the world create a healthy growth environment for high-temperature geothermal development. Geothermal well control needs to keep up with industry growth to ensure safe geothermal well operations and sustained geothermal energy delivery.

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Adapting Proven Oilfield Drilling Technologies to Deliver One of the Hottest Geothermal Wells Ever Drilled

by NOV

As global energy systems evolve toward lower-carbon, more resilient power generation, enhanced geothermal systems (EGSs) are attracting greater attention as a scalable source of reliable, continuous baseload power and a complement to intermittent renewables like solar and wind.

EGS projects aim to expand geothermal's geographic reach beyond traditional hydrothermal power systems, which are limited to locations where naturally occurring heat, water, and rock permeability are sufficient to extract geothermal energy. Instead, EGS developments target deep, hotter rock formations—generally greater than 3 km (1.9 miles) in depth with temperatures exceeding 150°C (302°F)—that lack natural permeability and existing water sources.

Accessing these formations at commercial scale pushes drilling systems beyond many high-pressure/high-temperature (HP/HT) drilling environments in oil and gas. High temperatures, abrasive rock, and complex well conditions place significant thermal and mechanical stress on drilling equipment and require greater operational precision than many traditional geothermal applications. But advances in drilling, stimulation, and fluid circulation techniques are helping EGS developers create artificial geothermal reservoirs capable of producing reliable, continuous geothermal energy for decades.

Service providers are leveraging application experience and engineering expertise to take modern drilling systems, many of which were first developed for complex oil and gas environments, and adapt them to meet the requirements of deeper, hotter geothermal environments.

At Fervo Energy's Project Blanford in Millard County, Utah, USA, a combination of digital drilling analytics, drill bit technologies, and completion solutions was used to support appraisal drilling operations.

Below, we examine the drilling technologies and operational approaches used to help efficiently drill the appraisal well under extreme thermal and geomechanical conditions.

Pushing the limits of conventional drilling technologies

High-temperature environments accelerate wear on drilling components and subject downhole tools to severe thermal cycling. At the same time, lithologies often consist of hard crystalline rock formations that can reduce drilling efficiency, increase mechanical loading on equipment, and shorten the life of drill bits and other downhole assets.

These conditions require both durable drilling hardware and precise control of drilling parameters to maintain operational stability. For Fervo, real-time drilling intelligence and drilling tools designed for extreme geothermal environments played an important role in enabling that control.

Real-time drilling intelligence in extreme conditions

One of the most critical elements in high-temperature geothermal drilling is maintaining visibility in drilling system performance. At Project Blanford, NOV provided a drilling optimization solution that captured high-resolution surface and downhole data throughout the drilling process, including:

- Weight on bit
- Rotary torque and RPM
- Rate of penetration (ROP)
- Standpipe pressure
- Pump rates and circulation flow

These measurements provided Fervo with a real-time view of mechanical and hydraulic conditions in the well.

Adapting Proven Oilfield Drilling Technologies to Deliver One of the Hottest Geothermal Wells Ever Drilled (Cont.)

by NOV

Engineers were able to quickly identify changes in drilling dynamics associated with all stresses and then adjust drilling parameters to help optimize bit performance, manage equipment stress, and reduce failure risks.

Fluid management is also critically important in high-temperature geothermal wells, where circulation conditions directly affect wellbore stability and the temperature of drilling components in the bottomhole assembly. A pit volume measurement system continuously monitored pit levels, flow-in and flow-out, and overall fluid behavior to provide additional hydraulic insights downhole. The system enabled early detection of potential circulation anomalies such as losses or influxes, helping maintain safe and stable drilling conditions throughout the operation.

To extend operational awareness beyond the rig site, a remote monitoring solution streamed drilling data to remote engineering and operations teams for real-time analysis and collaboration with field crews. This remote monitoring capability improved response times and supported faster operational decision-making as well conditions evolved. Together, these systems created an integrated drilling intelligence workflow capable of supporting complex geothermal drilling operations in extreme environments.

Drilling technologies engineered for hard, hot rock. While real-time data supported operational decision-making at Project Blanford, drilling performance also depended heavily on bit technologies capable of operating in hard, high-temperature formations.

Polycrystalline diamond compact (PDC) drill bits are increasingly being applied in geothermal drilling because their synthetic diamond cutters provide high wear resistance in abrasive geothermal formations.

For the ultra-high-temperature conditions encountered at Project Blanford, an optimized bit design was used, with advanced cutter technologies, modified blade geometry, and improved hydraulic flow management.

These engineering changes helped reduce mechanical stress on the bit while preserving cutter durability, contributing to higher ROPs, longer bit runs, and fewer trips out of the hole to replace worn bits.

Effective solids control is also important in EGS drilling, where hard formations generate large volumes of drilled solids or cuttings. At Project Blanford wellsite, shale shakers served as the primary solids-control stage, efficiently separating cuttings from the circulating fluid system to help recapture drilling fluids, maintain stable drilling fluid parameters, and support consistent drilling performance.

The shakers were designed to evenly deliver high G (gravitational) forces, a measure of the acceleration force applied to the drilling fluid and cuttings, across the width of the shaker screen basket. The shaker's 7G+ linear motion, coupled with an optimized fluid design to process fluids at rates between 450 and 750 gallons per minute, allowed for processing large fluid volumes while maximizing cuttings dryness and preserving drilling fluid. In controlled field testing, the shakers deliver solids removal efficiencies of up to 40%, which helps reduce unnecessary dilution, minimize waste volumes, and protect downstream equipment from excessive wear.

The shale shaker also incorporated an optimized vibration system that reduced operating noise compared with conventional designs, providing safer working conditions for rig crews while maintaining high solids-processing performance.

Adapting Proven Oilfield Drilling Technologies to Deliver One of the Hottest Geothermal Wells Ever Drilled (Cont.)

by NOV

Integrated drilling technologies support appraisal success

By adapting the suite of technologies described above for geothermal service, Fervo confirmed its geothermal resource potential at Project Blanford. PDC bits were deployed in both the intermediate and production sections of the appraisal well, including a single bit run that drilled 1,825 m (5,988 ft) in the intermediate interval. The same bit designs drilled most of the subsequent production interval, with PDC bits ultimately drilling approximately 78% of the total well footage.

Fervo drilled the appraisal well to a target depth of approximately 3.4 km (11,200 ft), where resource temperatures exceeded 290°C (555°F), surpassing the company's requirements for commercial viability. The well was drilled in less than 11 days, confirming the presence of an ultra-high-temperature geothermal resource with the potential to support multi-gigawatt development.

Applying oilfield drilling expertise to next-generation geothermal

Project Blanford highlights how drilling technologies originally developed for challenging oil and gas wells can be adapted for increasingly demanding geothermal applications.

As EGS projects target deeper and hotter resources, the ability to combine real-time drilling intelligence, high-performance drilling systems, and robust completion technologies will become increasingly important. Continued adaptation of established drilling technologies for geothermal service is expected to play a key role in improving drilling efficiency, extending equipment life, and supporting the commercial development of next-generation geothermal resources.



The Project Blanford project site in Millard County, Utah (source: Fervo Energy)

From Desktop Screening to Slimhole Confirmation: A Practical Workflow for Maturing EGS Prospects

by Kevin McCarthy

Global Geothermal Subject Matter Expert – Baker Hughes

Abstract

Many EGS prospects stall between promising desktop studies and the first confirmation well due to gaps in data integration and undefined decision criteria. This article presents a concise, six-stage workflow for advancing EGS prospects from regional screening to slimhole confirmation. It emphasizes clear decision gates, iterative physics-based temperature modeling, and structured data integration to reduce technical risk and optimize capital efficiency in early-stage EGS exploration.

Introduction and the Maturation Challenge

Play fairway analysis has advanced regional EGS prospect identification, yet efficiently advancing from desktop assessments to confidently drillable slimhole targets remains challenging. Projects often stall due to insufficient high-resolution data on fault architecture, stress regime, basement depth, and temperature distribution, or from poorly defined decision criteria and inadequate model updates. A structured workflow with clear decision gates bridges this gap, balancing technical rigor and capital efficiency based on lessons from successful and stalled EGS projects.

A Staged Workflow for Maturing EGS Prospects

The workflow comprises six practical stages, each with clear objectives, key activities, and decision gates to guide progression while managing risk and investment.

Stage 1: Desktop Screening & Resource Assessment

Regional play fairway analysis using gravity, magnetics, and existing datasets. Physics-based temperature modeling (variable thermal conductivity, temperature-dependent corrections) generates probabilistic temperature-depth profiles and MWe potential estimates. This serves as the primary technical-economic filter.

Decision Gate 1: Does the prospect show sufficient predicted temperature at economically drillable depths and viable thermal output to justify further geophysical investment and detailed modeling?

Stage 2: Geophysical Data Acquisition & Integration

Targeted MT, seismic (where appropriate), and high-resolution gravity/magnetic surveys. Integrate new data with well control and surface geology; update physics-based temperature models with improved structural and heat flow constraints.

Decision Gate 2: Does the integrated geophysical interpretation support a drillable target with acceptable risk?

From Desktop Screening to Slimhole Confirmation: A Practical Workflow for Maturing EGS Prospects (Contd.)

by Kevin McCarthy

Global Geothermal Subject Matter Expert – Baker Hughes

Stage 3: 3D Geological Modeling & Structural Framework

Construct 3D model integrating depth to basement, major fault architecture and damage zones, lithological variations affecting thermal conductivity and radiogenic heat production, and updated temperature distribution. Essential for meaningful well planning and risk assessment.

Decision Gate 3: Is the structural and temperature framework sufficiently constrained to support well targeting?

Stage 4: Well Targeting, Trajectory Design & Risk Assessment

Define well locations and trajectories that intersect favorable structural features, reach target temperatures, avoid major drilling hazards, and respect surface constraints (access, environmental, land). Incorporate preliminary 1D geomechanical analysis for fracture potential and stress implications.

Decision Gate 4: Are the proposed well locations and trajectories technically and economically justifiable?

Stage 5: Slimhole / Temperature Gradient Well Program

Drill slimhole exploration or temperature gradient wells for direct temperature measurement and model calibration, lithological and structural confirmation, stress and geomechanical data, and permeability/fluid chemistry information where possible. Rapidly integrate results to refine resource estimates before committing to larger-diameter wells.

Decision Gate 5: Do the slimhole results support progression to larger-diameter exploration or production wells?

Stage 6: Data Assimilation & Resource Model Update

Assimilate all new geophysical and slimhole data into updated models. Recalibrate temperature predictions, revise P10/P50/P90 resource size ranges, and reassess recovery factor assumptions. This stage closes the loop and prepares the prospect for development planning and larger-scale investment decisions.

The Role of Physics-Based Temperature Modeling

Throughout the workflow, physics-based adaptive temperature modeling serves as a living tool rather than a static exercise. As new data from geophysics and slimhole wells becomes available, models are iteratively updated. This provides more reliable temperature predictions than simple linear extrapolation and allows quantification of how new data reduces uncertainty in depth to target temperature and resource size estimates.

From Desktop Screening to Slimhole Confirmation: A Practical Workflow for Maturing EGS Prospects (Contd.)

by Kevin McCarthy

Global Geothermal Subject Matter Expert – Baker Hughes

Common Pitfalls That Cause Projects to Stall

Several recurring issues commonly delay or derail EGS prospects during maturation:

- Insufficient investment in 3D geological modeling before well planning
- Over-reliance on regional temperature maps without physics-based calibration
- Poor integration between geophysical interpretations and temperature modeling
- Lack of clear decision gates, leading to perpetual data collection without progression
- Underestimating the value of slimhole wells for model calibration before committing to larger wells

Conclusion (References)

Moving an EGS prospect from desktop screening to slimhole confirmation requires a disciplined, staged workflow with clear decision gates and continuous model updating. By integrating physics-based temperature modeling, focused geophysical acquisition, and 3D geological interpretation, project teams can significantly reduce technical risk and improve capital efficiency. This structured approach is particularly important as the geothermal industry moves beyond pure screening into active exploration drilling campaigns.

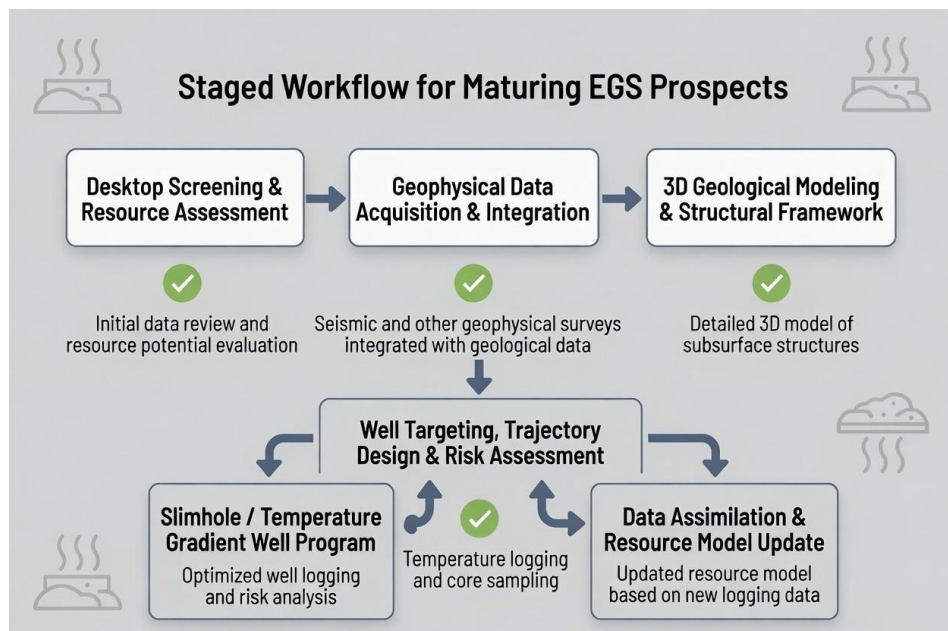


Figure: Overview of the staged workflow for maturing EGS prospects.

GEOHERMAL NEWS

OG Selects \$14 Million Enhanced Geothermal Systems Demonstration Project in Pennsylvania

Project will leverage oil and gas infrastructure to test enhanced geothermal systems in the eastern United States.

[U.S. Department of Energy's Office of Geothermal](#)

April 14, 2026

The U.S. Department of Energy's Office of Geothermal today announced a \$14 million project to support field tests for enhanced geothermal systems (EGS). Led by the Pennsylvania Department of Environmental Protection, the project will leverage the significant thermal resources in the Appalachian Utica Shale to assess the efficacy and scalability of EGS in the eastern United States.

The project activities include converting a horizontal shale gas well to geothermal. The effort will include assessing optimal well orientations and placements, as well as testing various techniques to create the fractures necessary for a human-made geothermal reservoir, or EGS. If successful, the project will offer a replicable model that could expand the use of EGS in more locations nationwide.

This project is the second selection under OG's EGS Pilot Demonstrations opportunity. The second-round of this opportunity was first announced in [June 2024](#). Learn more about the [EGS pilot projects](#) and OG's [other EGS research](#).

Read more here: [Link](#)

GEO THERMAL NEWS

EGEC hosts Policy Session at the European Sustainable Energy Week 2026

The European Geothermal Energy Council EGE C hosted a Policy Session on 9 June as part of the European Sustainable Energy Week (EUSEW) in Brussels. The Policy Session ‘*European Geothermal Action Plan for energy security, affordability and competitiveness*’ was prepared in close partnership with the European Commission (DG Energy) and Canada’s Mission to the EU.



Read more: [Link](#)

According to EGE C, the Commission’s proposal must include the following elements:



- **An EU-wide target to reach 250 GW of geothermal capacity by 2040** covering all technologies. This will build on the 44 GW of installed capacity in the EU.
- **A European Geothermal Charter**, which codifies the 2040 target and launches the **Geothermal Industrial Alliance** to manage its delivery.
- Targeted **European financial instruments** to leverage private capital.
- Measures to make **permitting processes** more efficient and faster, as well as improving **access to geological data**.
- **Sectoral or tripartite agreements** with key energy consumers.
- Peer-to-peer guidance to help governments develop **national and regional roadmaps** to remove barriers, accelerate investments and build local supply chains.
- European instruments to support **value chain development** in local manufacturing and skilled professionals.
- The inclusion of geothermal energy as a central pillar of the EU’s **Global Gateway** and the **Global Energy Transition Forum**.
- Improvements to the collection and presentation of **market data and statistics** to enable better energy modelling.

GEOHERMAL NEWS

Mantle Energy raises \$5m seed funding to generate geothermal energy in oilfields

[Hunt Energy Company](#) (Hunt Energy) has announced that its subsidiary, [Mantle Energy](#), has closed a \$5 million seed round led by 17Shoals Inc. The seed funding will go towards scaling pilot projects, securing key partnerships, and laying the foundation for future as the company prepares for a Series A funding raise.

Mantle Energy seeks to utilize the remaining potential in shale formations, corresponding to about 90% of the hydrocarbons that remain in the formation after oil production. The proposed concept is to heat the underground through controlled combustion. This heat is then captured to produce electricity. The concept is novel and yet again challenges what is typically defined as 'geothermal energy.'

Read more: [Link](#)

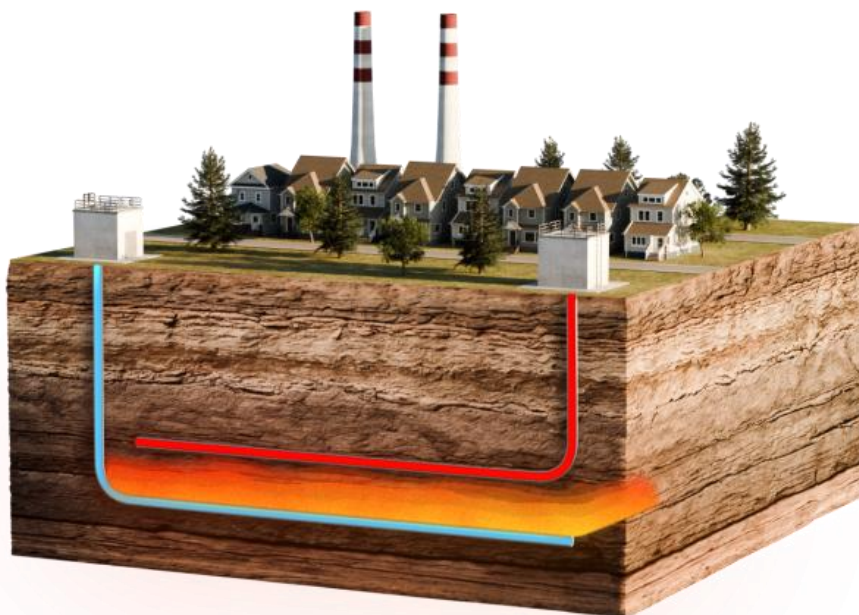


Fig: Diagram of Mantle Energy's proposed method of producing geothermal energy in oilfields (Source: Mantle Energy)

SPE GTTS ACTIVITIES: WEBINARS

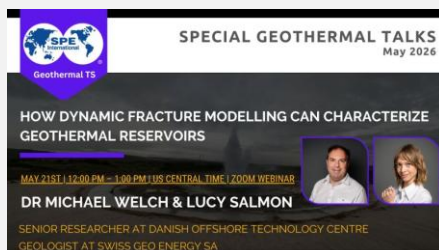
Why Geothermal Testing Needs a Different View: An Experimentalist Opinion



 **Recording link:** [SPEcial GTTS April 2026 webinar: Why Geothermal Testing Needs a Different View: An Experimentalist Opinion](#)

In this webinar **Dr. Catalin Teodoriu**, Professor of Petroleum Engineering at the University of Oklahoma, will challenge the status quo with insights from one of the world's most advanced drilling & geothermal research labs.

How Dynamic Fracture Modelling Can Characterize Geothermal Reservoirs



 **Recording link:** [SPEcial GTTS May 2026 webinar: How dynamic fracture modelling can characterize geothermal reservoirs](#)

In this webinar, Dr. **Michael Welch** (DTU - Technical University of Denmark) and **Lucy Salmon** (Swiss Geo Energy SA) shared about their physics-based forward modelling tool that simulates fracture network growth using Linear Elastic Fracture Mechanics and Subcritical Propagation Theory. As part of the EU Horizon-funded Go-FORWARD project, they are adapting this for geothermal targets like tight carbonates and crystalline basement rocks.

The Role of the Geosciences in the Energy Transition



 **Recording link:** [SPEGeothermal June 2026 webinar: The Role of the Geosciences in the Energy Transition](#)

This webinar highlights how geoscientists as "subsurface integrators," combine geological, geophysical, and geochemical data to build integrated reservoir models that significantly de-risk geothermal exploration.

SPE GTTS ACTIVITIES: CONFERENCES

Building the Bridge Oil and gas expertise for Geothermal – SPE Italian Section Milan, Italia. April 16, 2026

On 16 April 2026, Princewill Okechukwu represented the SPE Geothermal Technical Section (SPE GTTS) in Milan at Building the Bridge: Oil & Gas Expertise for Geothermal Development, a one-day event hosted at the Eni Bolgiano Auditorium. The event was conceived as a high-quality platform for policy-industry dialogue, networking, and knowledge-sharing, with a specific focus on how oil and gas capabilities can be leveraged and repurposed to accelerate geothermal development. The invited audience included industry practitioners, developers, utilities, technology providers, policymakers, and researchers from across Europe.

Princewill delivered the opening remarks. The presentation introduced the section, highlighted its active engagement within the industry, emphasized the strong synergies between the oil & gas and geothermal sectors, and encouraged attendees to join and participate in GTTS activities.

Expertise for Geothermal Development

During the opening presentation, Princewill Okechukwu focused on a central message: geothermal energy should no longer be viewed as a niche technology operating in isolation. Instead, the sector increasingly requires industrial-scale development, faster deployment, and stronger execution capabilities. He emphasized that many of the competencies needed to accelerate geothermal growth already exist within the oil and gas industry, including drilling, completions, reservoir engineering, subsurface analytics, and field operations. Cross-sector knowledge transfer was presented not as a theoretical concept, but as a practical pathway to reduce risk and accelerate the commercialization and deployment of geothermal projects.

A key theme throughout the presentation was the value that oil and gas expertise can bring beyond technical capabilities alone. Princewill highlighted how subsurface knowledge contributes to improved resource characterization and risk reduction, while drilling and completions experience supports the development of deeper, hotter, and more complex geothermal wells. Reservoir engineering practices enhance sustainable production planning, and the industry's project discipline and execution culture can help geothermal move beyond pilot-stage initiatives toward scalable and repeatable deployment. This message resonated strongly with the audience, which included a broad mix of energy professionals, engineers, and decision-makers familiar with subsurface industries.



SPE GTTS ACTIVITIES: CONFERENCES

Building the Bridge Oil and gas expertise for Geothermal – SPE Italian Section Milan, Italia. April 16, 2026

The presentation also provided an opportunity to introduce SPE's Geothermal Technical Section (GTTS) as a global technical forum specifically positioned to support this transition. GTTS was presented as a platform for sharing geothermal knowledge, facilitating technology transfer from oil and gas to geothermal development, and creating opportunities for collaboration, professional growth, and business development across regions. Attendees who were already SPE members were encouraged to join the section to remain connected with the latest technical developments, webinars, and achievements across the geothermal community.

The remainder of the event reinforced the significance of the gathering beyond its symbolic value. The conference brought together key stakeholders from the Italian and wider European geothermal and energy ecosystem, including representatives from industry, academia, research organizations, and policy institutions. Discussions provided a realistic and constructive assessment of the current state of geothermal development in Italy and Europe, covering the importance of supportive policy frameworks, new exploration activity, the improving competitiveness of geothermal energy, the engineering challenges associated with drilling and surface facilities, and the need to translate ambition into implementation. Contributions from Project InnerSpace, particularly regarding geothermal databases and mapping initiatives, highlighted the importance of improving access to structured subsurface data and strengthening collaboration across the energy industry.

From the GTTS perspective, the event was considered highly successful. It provided an opportunity to position SPE GTTS as the global platform connecting oil and gas expertise with geothermal opportunity, while helping establish the context for many of the discussions that followed. A key takeaway emerging from the event was that geothermal energy is no longer on the margins of the energy transition. The technological foundations are already in place, the industrial rationale is increasingly compelling, and the urgency to expand deployment continues to grow. The next challenge is achieving stronger alignment among stakeholders, securing greater investment, and accelerating action at scale. In this context, GTTS has an important role to play in facilitating knowledge exchange and supporting the industry's continued development.



SPE GTTS ACTIVITIES: CONFERENCES

SPETech Days
Bucharest, Romania. May 10-12, 2026

SPE Geothermal Technical Section - GTTS Section (GTTS) board member had the privilege of representing the section at the SPE Tech Days in Romania organized by **SPE Romanian Section**, and what an experience it was.

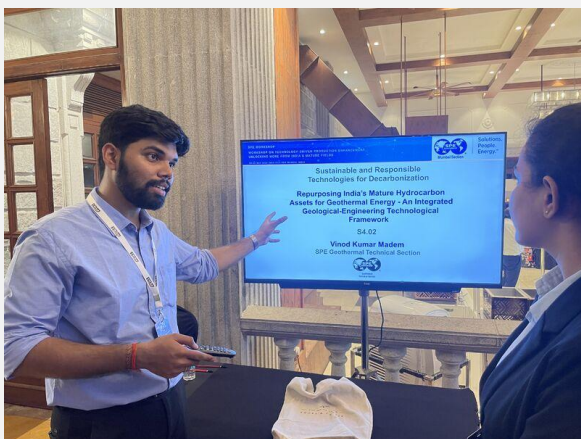
Our community is growing and fast. Every conversation at events like this reminds us why the SPE Geothermal Technical Section - GTTS exists: to be the bridge between the technical depth of oil and gas and the enormous promise of geothermal energy.



SPE GTTS ACTIVITIES: CONFERENCES

Workshop on Technology-Driven Production Enhancement: Unlocking More from India's Mature Fields Mumbai, India. May 22-23, 2026

SPE Mumbai Section has hosted this workshop where our India Membership Chair **Vinod Kumar Madem** contributed and represented the SPE Geothermal Technical Section. He engaged with industry leaders, discussing the future of our energy landscape!



Vinod presented a poster titled "Repurposing India's Mature Hydrocarbon Assets for Geothermal Energy – An Integrated Geological-Engineering Technological Framework", in the "Sustainable and Responsible Technologies for Decarbonization" session where he explored how we can strategically pivot our existing mature fields toward sustainable geothermal energy extraction. By leveraging existing infrastructure in areas like the Pranhita-Godavari basin and other mature hydrocarbon candidates across India, India can significantly accelerate its renewable energy goals by harnessing un-tapped geothermal energy.



SPE GTTS ACTIVITIES: CONFERENCES

Utah FORGE Geothermal Workshop Salt Lake City, UT. June 1-3, 2026

The SPE Geothermal Technical Section participated in the University of Utah Geothermal Workshop, hosted by the Energy & Geoscience Institute and Utah FORGE — the U.S. Department of Energy's dedicated EGS field laboratory. The three-day event in Salt Lake City brought together researchers, operators, and service companies to share insights on enhanced geothermal systems development.

The workshop featured Utah FORGE team presentations on recent geologic, geochemical, and geophysical work, alongside expert panels on R&D outcomes, operator challenges, and service company innovation. Breakout sessions covered EGS stimulation design, monitoring strategies, and field development optimization.

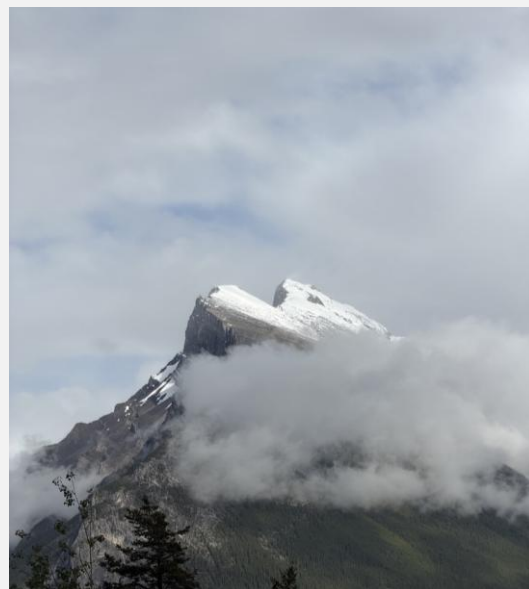


SPE GTTS ACTIVITIES: CONFERENCES

Superhot Rock Summit Banff, Canada. June 5-7, 2026

The SPE Geothermal Technical Section participated in the Superhot Rock Geothermal Summit, co-organized by the Clean Air Task Force (CATF) and the Bezos Earth Fund. The summit brought together geothermal innovators, researchers, investors, and policymakers in Banff, Canada to advance superhot rock geothermal energy.

The summit featured expert panels on superhot rock field projects and hard problems facing the industry. Working sessions on drilling challenges, heat extraction, and commercial deployment strategies produced technology roadmaps on bit design and insulation systems.



SPE GTTS ACTIVITIES: CONFERENCES

World Geothermal Congress (WGC) Calgary, Canada. June 8-11, 2026

The SPE Geothermal Technical Section (GTTS), proudly took part in the recent World Geothermal Workshop WGC, in Calgary. Our participation reinforced the value of staying actively engaged with the broader geothermal community, where the exchange of ideas and continuous collaboration are essential to advancing the sector. The workshop offered an excellent platform to reconnect with long-standing colleagues, establish new connections, and gain fresh perspectives through insightful presentations and open technical discussions. Strengthening these relationships is key to driving innovation and ensuring we remain at the forefront of geothermal development.



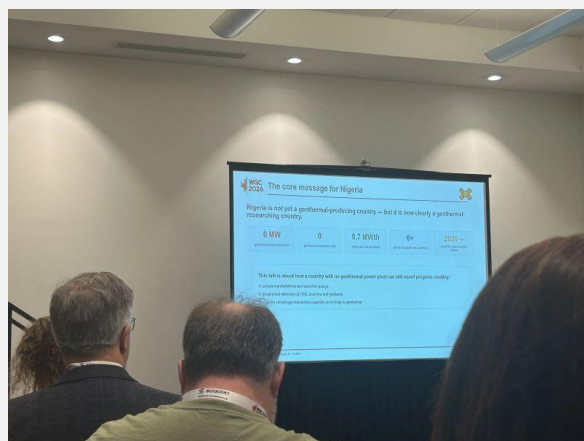
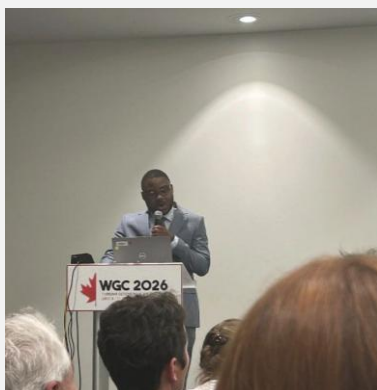
SPE GTTS ACTIVITIES: CONFERENCES

World Geothermal Congress (WGC) Calgary, Canada. June 8-11, 2026

Princewill Okechukwu presented Nigeria's humble Geothermal Country Update at the ongoing **World Geothermal Congress 2026** in Calgary, Canada.

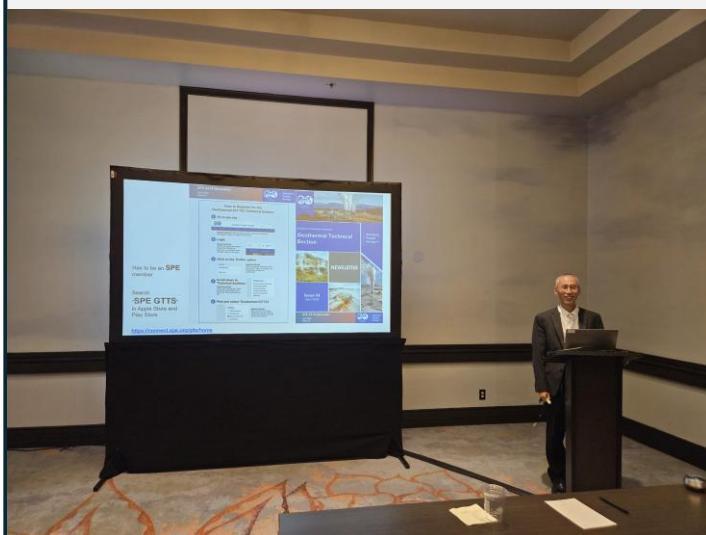
---"Nigeria does not yet have an operational geothermal power plant, production well, or active geothermal field development project. But that does not mean there is no progress. Our update focused on the growing research evidence base across Nigeria, from Ikogosi to Ruwan-Zafi to the Benue Trough, and other structurally favourable areas. Showing that the country has moved from geothermal curiosity to a stronger scientific foundation for future exploration was worth the trip".

The key message was simple: Nigeria is not yet a geothermal-producing country, but it is now clearly a geothermal-researching country — and the next step must be visible field action. For Africa's largest oil producer, Princewill also emphasized the opportunity to leverage existing oil and gas expertise, legacy subsurface data, drilling knowledge, reservoir understanding, and field operations experience as a bridge toward geothermal exploration and development.



SPE GTTS ACTIVITIES: CONFERENCES

The 60th US Rock Mechanics / Geomechanics Symposium of the American Rock Mechanics Association
Tucson, Arizona, USA, June 21-24, 2026



Gary Chen and Wei Fu led the ***“Repurposing Existing Oil and Gas Wells for Geothermal Applications”*** workshop at the 60th ARMA Symposium. Hats off to all the presenters for sharing their work and insights throughout the workshop. Their contributions made the discussions practical, technically enriching, and truly inspiring.

From AI/ML-driven large-scale well filtering to data-based candidate screening, from well construction and integrity to open-loop and closed-loop geothermal applications, from evaluation workflows to solid case studies, and from geothermal hot/cold energy storage to district heating, power generation, and annulus-circulating co-production concepts, the workshop highlighted a new wave of momentum and opportunity in repurposing oil and gas wells for geothermal energy.



SPE GTTS ACTIVITIES: CONFERENCES

SPE Europe Energy Conference & Exhibition ISTANBUL, TÜRKİYE. June 23-25, 2026

Advancing the Energy Transition: SPE GTTS at EuropEC 2026

The **SPE Europe Energy Conference and Exhibition (EuropEC 2026)**, held this past June in Istanbul, Türkiye, served as a premier collaborative stage for the global energy transition. A key highlight for our community was the active participation of **Gabrijel Grubac**, Vice-Chair of the **SPE Geothermal Technical Section (GTTS)**.

Bridging Subsurface Expertise

Centering on the event's core focus of cross-disciplinary innovation, Gabrijel spearheaded engagements at **Booth A01 (SPE Technical Sections)**. Alongside fellow global section leaders, he connected with operators, researchers, and young professionals to highlight how traditional petroleum engineering skill sets—such as reservoir modeling and deep subsurface analytics—are directly accelerating the deployment of geothermal energy systems and Enhanced Geothermal Systems (EGS).



SPE Upcoming Conferences

SPE Annual Technical Conference and Exhibition (ATCE)

Houston, Texas, USA. October 21-23, 2026

SPE  ATCE

Link to conference:
<https://www.atce.org/>

21–23 OCTOBER 2026

GEORGE R. BROWN CONVENTION CENTER

HOUSTON, TX, USA

Geothermal - Technical Session:

11 Next-Generation Geothermal Systems: Materials, Modeling, and Performance

Thursday, 22 October

Room 361 AD | 0830 – 1130

This session focuses on advanced modeling and performance optimization of geothermal systems, with emphasis on Enhanced Geothermal Systems (EGS). Topics include coupled thermo-hydro-mechanical modelin ...

Session Chairpersons

-  Fatick Nath - Texas A&M International University
-  Cassandra Dewan - SLB
-  Joseph Spencer - Baker Hughes
-  Zhenzhen Wang - Chevron Corporation

16 Innovative Techniques and Technologies for Well Integrity Across Geothermal, Abandonment and CCS Operations

Thursday, 22 October

Room 371 AD | 0830 – 1130

This session showcases innovative techniques and technologies advancing well integrity across geothermal, CCS, and well abandonment applications. Presentations cover cutting edge developments such as ...

Session Chairpersons

-  Peter Janiczek - OMV E&P GmbH
-  Albert McSpadden - Altus Well Experts Inc.
-  Sulaiman Sidek - PETRONAS Carigali Sdn Bhd

38 Leveraging Technical Excellence and Innovation for Complex Wells: Geothermal, CCUS and MPD

Room 371 CF | 0830 – 1130

Friday, 23 October

As the industry expands beyond conventional oil and gas, established drilling expertise combined with innovation in research and technologies are being leveraged to address new challenges, improve per ...

Session Chairpersons

-  Romar Gonzalez-Luis - Blade Energy Partners
-  Isabel Poletzky - Premium Oilfield Technologies
-  Mohamed Eishabrawy - XGS Energy

GEOHERMAL CONFERENCES

Click on the title of each conference to get more information

July, 2026

15th ITB International Geothermal Workshop - July 13 - 14, 2026 - Bandung, Indonesia

SPE Africa Geothermal Workshop - July 20 - 21, 2026 - Nairobi, Kenya



August, 2026

NZ Geothermal Week - August 10 - 15, 2026 - Taupo, New Zealand

12th Indonesia International Geothermal Convention & Exhibition - August 19 - 21, 2026 - Jakarta, Indonesia

September, 2026

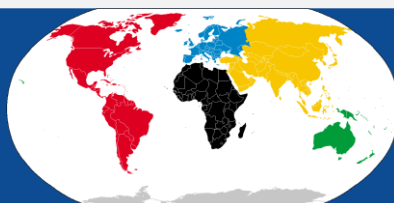
7th Philippine International Geothermal Conference - September 2 - 4, 2026 - Legazpi, Philippines

Geothermal District Heating & Cooling Days - September 3 - 4, 2026 - Dublin, Ireland

Budapest Geothermal Energy Summit - September 14, 2026 - Budapest, Hungary

Celle Drilling 2026 - September 15 - 16, 2026 - Celle, Germany

Geothermal Rising Conference - September 20 - 23, 2026 - Houston, Texas



GEOTHERMAL CONFERENCES (Cont.)

Click on the title of each conference to get more information

October, 2026

Praxisforum Tiefen Geothermie - October 14 - 16, 2026 - Pullach, Germany

11th African Rift Geothermal Conference (ARGeo C11) - October 19 - 24, 2026 - Moroni, Comoros

German Geothermal Congress / Der Geothermie Kongress - October 20 - 22, 2026 - Potsdam, Germany

November, 2026

International Geothermal Investment Summit - November 2026 - Cappadocia, Türkiye

EAGE Global Energy Transition - November 2 - 6, 2026 - Hannover, Germany

13th UK Geothermal Symposium - November 9 - 11, 2026 - Leeds, UK

2nd EAGE / AAPG Geothermal Energy in the Middle East Workshop - November 9 - 11, 2026 - Jeddah, Saudi Arabia

IADC Geothermal Drilling Conference & Exhibition - November 17- 18, 2026 - Copenhagen, Denmark

48th New Zealand Geothermal Workshop - November 18 - 20, 2026 - Auckland, New Zealand

CONNECT 2026 - November 19, 2026 - Bern, Switzerland

December, 2026

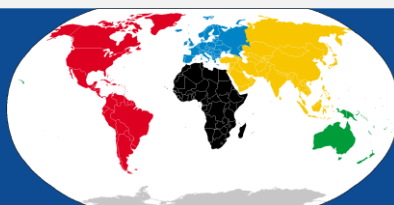
SPE Forum: Unlocking the Earth - Future-Forward Strategies for Scalable Geothermal - December 8 - 11, 2026 - Galveston TX

February, 2027

GeoTHERM expo & congress - February 25 - 26, 2027 - Offenburg, Germany

March, 2027

2nd Geo-RIN Conference - March 9 - 12, 2027 - São Miguel, Azores, Portugal



SPE Geothermal Technical Section ATCE 2026 Networking Event



The countdown has officially begun for the premier event on the Society of Petroleum Engineers' calendar. From **21–23 October 2026**, the global energy community will converge at the George R. Brown Convention Center in **Houston, Texas**, for the **SPE Annual Technical Conference and Exhibition (ATCE 2026)**.

Under this year's banner, *"Exploring New Horizons with Innovative Solutions,"* ATCE is sharpening its focus on cross-discipline synergy and the rapid maturation of alternative energy workflows. Positioned at the forefront of this movement, the **SPE Geothermal Technical Section (GTTS)** invites all members, operators, and young professionals to join them for a series of high-impact engagements, capped off by the highly anticipated **GTTS Annual Networking Event** and get an exclusive chance to:

- **Meet the Leadership:** Connect directly with the GTTS Board members and sub-committee leads to share your ideas and learn about upcoming technical initiatives, specialized working groups, and global knowledge-sharing platforms.
- **Forge New Collaborations:** Build cross-functional partnerships with operators, technology service providers, and academic institutions actively investing in deep-heat applications.

Secure Your Spot: Registration and Planning

- **Dates:** 21–23 October 2026
- **Location:** George R. Brown Convention Center | Houston, Texas, USA
- Link: [Registration](#)

Don't miss out on the conversation shaping the future of subsurface heat. Make sure your calendar is marked for the GTTS reception, and let's keep driving innovation beyond traditional reservoirs together!

COLLABORATION

SPE Africa Geothermal Workshop July 20th-21st, 2026

Powering Africa's Energy Future with Geothermal Innovation

The SPE Geothermal Technical Section is proud to jointly host the SPE Africa Geothermal Workshop 2026 with SPE Africa Region, taking place in Nairobi, Kenya, the heart of Africa's geothermal excellence.

Under the theme "Geothermal for Africa's Energy Future," this two-day workshop will bring together policymakers, financiers, operators, researchers, and innovators to explore how geothermal can become a pillar of Africa's clean energy transition



The graphic features a blue and yellow color scheme. At the top, it displays the SPE Africa Region and SPE Geothermal Technical Section logos alongside the tagline "Solutions. People. Energy.™". The main title "SPE AFRICA GEOTHERMAL WORKSHOP 2026" is prominently displayed in blue and yellow. Below this, the theme "Harnessing Africa's Geothermal Potential for a Sustainable Energy Future" is highlighted. The event dates "20th - 21st JULY 2026" and location "Fairmont The Norfolk Hotel NAIROBI, KENYA" are clearly stated. A "REGISTER HERE" link is provided, along with the hashtag #SPEAFRICAGEOTHERMALWORKSHOP and social media icons for LinkedIn, Facebook, Instagram, and X.

**SPE AFRICA
GEOTHERMAL
WORKSHOP 2026**

THEME
Harnessing Africa's Geothermal
Potential for a Sustainable Energy Future

20th - 21st Fairmont The Norfolk Hotel
JULY 2026 NAIROBI, KENYA

REGISTER HERE
<https://speafricageothermalworkshop.eventbrite.com>

#SPEAFRICAGEOTHERMALWORKSHOP

in f @ x SPE Africa Region

COLLABORATION

SPE Africa Launches Geothermal Datathon

July 20th-21st, 2026

Igniting Innovation Across Data Science and Geothermal Energy

With development accelerating across Africa, the continent's energy demand is growing at a remarkable rate of 4.5% annually. Despite abundant energy resources, many remain under-utilised. Geothermal energy stands out as a promising source, with significant progress especially in Eastern Africa where the Rift Valley stretches from Eritrea to Mozambique. Currently, geothermal energy comprises only a small share of Africa's energy mix, though in Kenya it supplies nearly **28–50%** of national electricity. The region's potential is estimated at **15,000–20,000 MW** electricity.

To harness this potential, **SPE GTTS**, **SPE DSEATS Africa Region**, and **SPE Kenya Section** have organised the inaugural SPE Africa Geothermal Datathon. This event aims to equip geothermal professionals and inspire interest in the intersection of data science and geothermal development.

The need for such a program has been evident from the number of registered participants who are over 250 spanning across the continent and beyond. Clearly, steam has been building up and we found a way to harness it. This provides an opportunity to the participants to channel momentum into practical solutions for the sector.

This hackathon challenges participants to design an integrated geothermal heating and cooling system for a mixed-use urban cities, inspired by real geothermal developments. Using subsurface data from several existing wells, teams will assess whether geothermal energy can reliably supply a neighbourhood demand of at least 10 MW of heating and 5 MW of cooling. Beyond evaluating the geothermal resource, participants must develop an innovative energy system concept that combines geothermal production with surface technologies such as heat pumps, chillers, thermal storage, and hybrid renewable solutions. The challenge also encourages the use of public datasets and AI-assisted workflows to improve resource assessment, optimize system design, and evaluate economic performance, providing a hands-on opportunity to tackle real-world sustainable energy challenges.

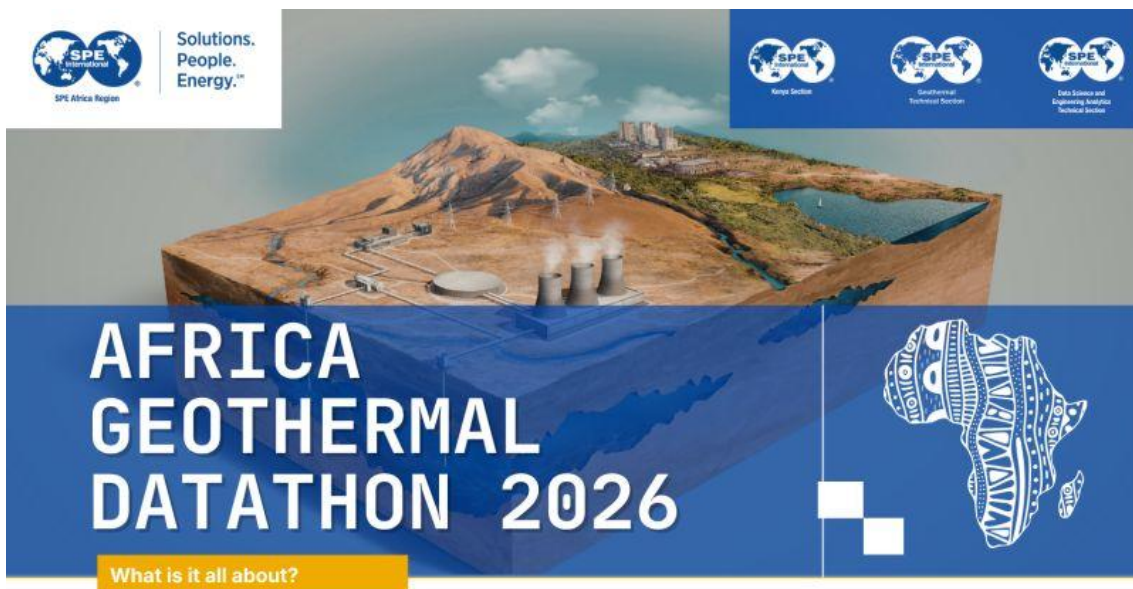
The success of the first edition is thanks to the dedication of the volunteer organising team:

- **SPE GTTS:** Pejman Shoeibi Omrani, Princewill Okechukwu, Katerina Yared, Gabriel Grubac
- **DSEATS Africa Region:** Michael Aku, Michelle Boit, Anita Badagawa
- **SPE Kenya Section:** Sarah Ali, Arthur Kisaka, Peter Karu

These individuals have been crucial in developing the challenge, creating a competition framework, marketing, engaging participants, and reviewing solutions objectively.

All eyes now turn to Nairobi as top teams present their solutions at the **SPE Africa Annual Student Congress (SAASC)**, held 22nd–24th July 2026.

Is this the last one? Definitely not, we are committed to ensuring that the SPE Africa Geothermal Datathon remains a part of the annual calendar. We hope that with subsequent editions we will be able to secure data partners, sponsors and much more in terms of awards to participants.



The Africa Geothermal Datathon is a competition designed around solving challenges in the African geothermal industry through data science, engineering analytics, machine learning, automation and AI. It is aimed at enabling students and young professionals to grow, test and gauge their understanding while exposing them to issues relevant to industry

What to expect

Participants will be required to form teams of 3 - 5 members then provided with a challenge, a dataset and relevant learning resources.

Based on the dataset and their expertise, they will be required to find a unique and creative solution to the provided challenge within the provided time

Once your submission is made, it will be reviewed and ranked by the judges.

The top teams will present their solutions during the SPE Africa Annual Students Congress in Nairobi, Kenya with subsequent awards.

Important Dates

- Kick Off Webinar
May 11, 2026
- Submission Deadline
June 4, 2026
- Presentations & Awards
July 24, 2026

Who its for

STUDENTS

Students interested in the intersection of data and geothermal energy systems

YOUNG PROFESSIONALS

Young professionals shaping Africa's geothermal energy, geological knowledge and data landscape .

Are you ready?

An opportunity to learn, grow, compete and win !!!

Register via

<https://dub.sh/africadatathon26>

Contact Us

speafricageodatathon@gmail.com

COLLABORATION OPPORTUNITIES

Expanding GTTS Impact through Cross-sectional Engagement

The **GTTS Technical Directors team** is preparing to expand its engagement across SPE by strengthening collaboration with other Technical Sections. In the coming months, we aim to **leverage cross-disciplinary synergies, co-develop impactful events, and further position geothermal within the broader energy ecosystem.**

We invite all SPE Technical Sections interested in joint initiatives—whether workshops, webinars, or technical exchanges—to connect with us as we roll out this effort. GTTS is committed to building strong, collaborative partnerships that drive innovation and knowledge sharing across the industry.

If your section is interested in engaging, we welcome the conversation and look forward to working together.

Kindly contact: **Vinod Kumar Madem** (Membership Chair, India)



Technical Sections

- Artificial Lift - Asset Management - Carbon Dioxide Capture, Utilization and Storage - Critical Minerals - Data Science and Engineering - Drilling Systems Advancement - Drilling Uncertainty Prediction - Flow Assurance - Flow Measurement - Geomechanics - Geothermal - Human Factors - Hydraulic Fracturing	- Hydrogen - Integrated Reservoir Management - Methane - Offshore Completions - Plug and Abandonment - Research and Development - Robotics and Autonomous Systems - Separations Technology - Sustainable Development - Water Life Cycle and Strategy - Wellbore Positioning - Well Integrity
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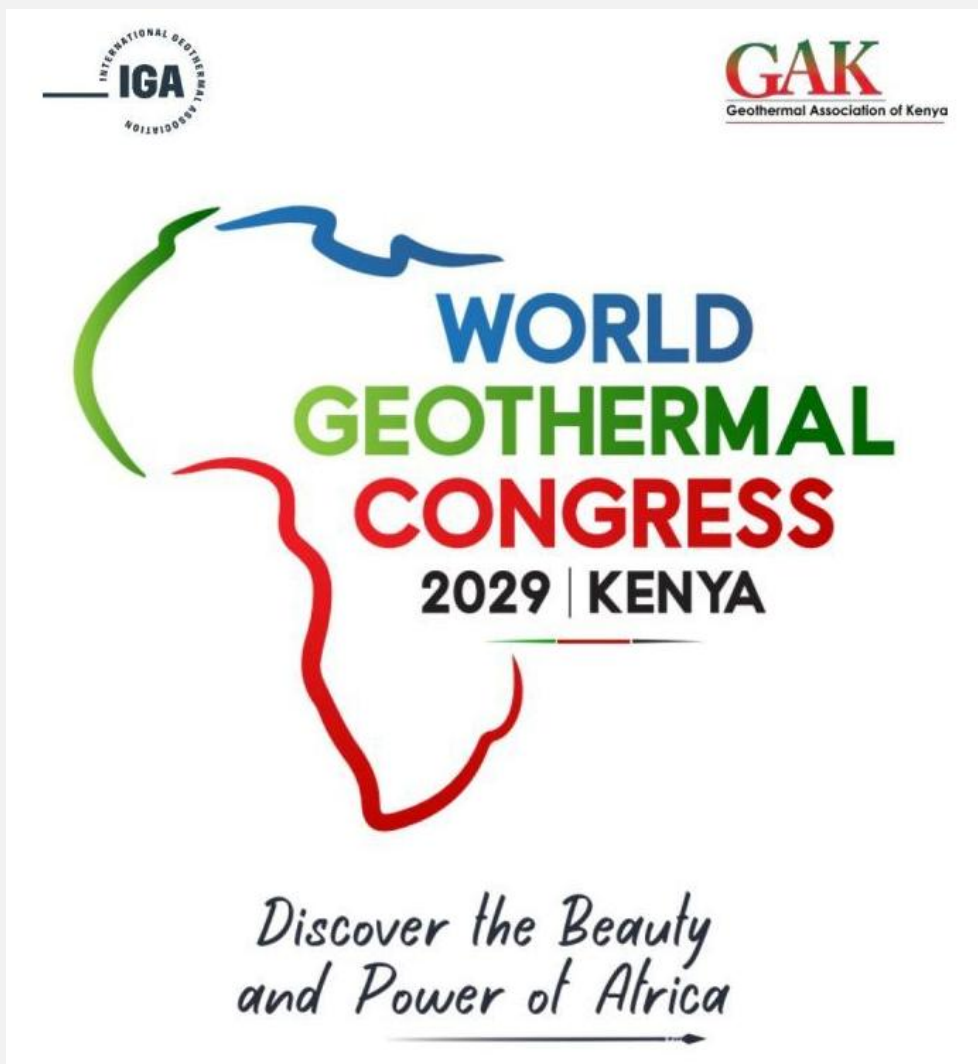
GEOHERMAL CONFERENCES

Great news for geothermal energy in Africa!. **Kenya is selected to host the World Geothermal Congress 2029.**

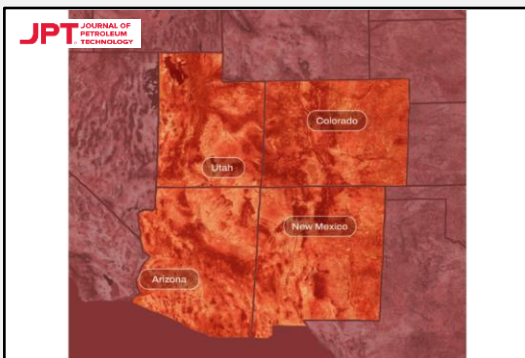
Nairobi, Kenya has been selected to host the World Geothermal Congress 2029, the first time this global event will be held on the continent. A huge milestone for the region and a strong recognition of Africa's leadership in geothermal development.

Congratulations to International Geothermal Association and Geothermal Association of Kenya (GAK) for spearheading it and enabling an increase in geothermal momentum.

Press Release



FEATURED JPT GEOTHERMAL ARTICLES



Four Western US States Form Consortium To Advance Geothermal Energy Development

Utah, Colorado, Arizona, and New Mexico have formed the Mountain West Geothermal Consortium to turn geothermal energy into gigawatts of baseload power.

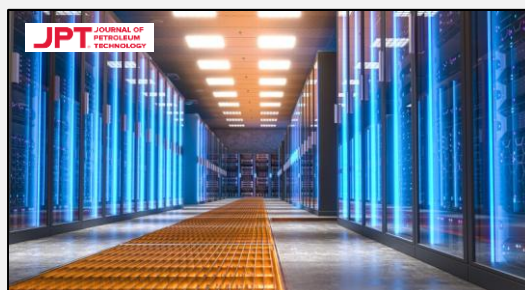
June 5, 2026 By JPT Staff
Journal of Petroleum Technology



Baker Hughes, H&P Team Up on Geothermal Development

The collaboration aims to provide a dedicated land rig for geothermal wells in the US.

May 24, 2026 By JPT Staff
Journal of Petroleum Technology



AI Data Center Growth Pushes Enhanced Geothermal Into the Energy Spotlight

AI-driven data center growth is straining US power grids and accelerating interest in enhanced geothermal systems as a scalable, low-carbon solution.

May 6, 2026 By Pat Davis Szymczak
Journal of Petroleum Technology



Fervo Energy and Vallourec Sign 5-Year Deal To Scale US Geothermal Projects

Fervo Energy and Vallourec have signed a 5-year supply agreement to support large-scale deployment of next-generation geothermal across the US, establishing a fully domestic supply chain for geothermal well components.

April 22, 2026 By JPT Staff
Journal of Petroleum Technology



Geothermal Firm Fervo Energy Debuts on US Stock Market

The Houston-based enhanced geothermal developer scored \$1.9 billion in an initial public offering, positioning it to expand projects in Utah and Nevada.

May 14, 2026 By JPT Staff
Journal of Petroleum Technology

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IADC Releases Geothermal Well-Control Guidelines

JPT JOURNAL OF
PETROLEUM
TECHNOLOGY

The new guidelines were released prior to reports that enhanced geothermal developer Fervo Energy experienced a blowout in Utah with no reported injuries.

June 9, 2026
BY JPT STAFF
Journal of Petroleum Technology

The [International Association of Drilling Contractors \(IADC\)](#) has released its first guidelines for [geothermal](#) well control, outlining best practices for drilling, workover, and well-service operations. The guidelines, [available here](#), apply to geothermal projects ranging from subhydrostatic conditions to wells capable of unassisted flow and are intended to protect workers, the public, and the environment.

The IADC describes the document as a consistent framework that applies to site procedures, equipment testing, operational planning, and personnel training. Additional topics include barrier philosophy, drilling, completion, and permanent abandonment. While the document excludes nonstandard well-control operations, it provides recommended safety protocols and warning indicators for geothermal conditions involving steam or high-pressure fluids.

The guidelines note that a geothermal well may be considered “live” due to both elevated formation pressure and high temperature. It adds that a reduction in hydrostatic pressure can allow superheated water to flash into steam, resulting in rapid expansion that may lead to a blowout.

The document also highlights that [geothermal well-control practices](#) are adapted from conventional oil and gas well control but modified to address the unique challenges of geothermal environments. “A thorough understanding of subsurface reservoir behavior, fluid properties, flash potential, in-situ pressure, and temperature conditions is essential to shaping an effective well-control strategy. In most geothermal scenarios, both pressure and temperature are critical factors,” the guidelines state.

The release follows IADC’s 2025 introduction of its [Geothermal Well Classification](#) framework, which seeks to standardize classification to improve efficiency, safety, and cross-sector collaboration.

The publication of the geothermal well-control guidelines came only days before reports surfaced that [Fervo Energy](#) experienced a blowout at its Cape Station project in Beaver County, Utah. According to reports from Axios and other outlets, the incident was contained by the geothermal energy developer as of 29 May, with no injuries or environmental damage reported.

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EDITORIAL TEAM MESSAGE

Synergies in the Subsurface – Geothermal Developments Across the World

Fifth Edition of the SPE GTTS Newsletter

Welcome to the fifth issue of the SPE Geothermal Technical Section Newsletter!

In this edition, we turn our attention to a powerful subsurface alliance gaining momentum globally: the intersection of Geothermal Energy and CCUS. Across Europe, North America, and Asia-Pacific, operators and researchers are unlocking breakthroughs by utilizing captured CO₂ as a working fluid in geothermal systems (CO₂-EGS). This approach not only enhances heat extraction efficiency but simultaneously achieves permanent geological storage.

Repurposing mature sedimentary oil and gas basins for geothermal heat extraction represents a dual-win framework for the energy transition. It allows our industry to leverage existing asset infrastructure, rich legacy datasets, and reservoir engineering expertise to advance decarbonization while generating sustainable baseload power.

As a global technical community, exploring these hybrid subsurface technologies is vital to driving both economic viability and net-zero goals forward. We hope this issue inspires fresh collaborative ideas across both domains.

Enjoy reading!

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SPEcial Geothermal Workshop Europe - Loading...

SPE Geothermal Technical Section is working on a geothermal workshop in Europe. Coming in November 2026!

Stay tuned as we share more details in the coming weeks....

INVITING ARTICLES FOR THE NEXT ISSUE

(Deadline: 31st August, 2026)

We are now inviting articles for our next issue. Whether you are working on a breakthrough research paper, a field case study, or a perspective on the latest regulatory changes in the geothermal space, your voice matters. Below are the guidelines for submission, and the article size is limited up to 2 pages.

- The content must have real educational or technical value on its own.
- Do not mention a company's name more than is needed to give them credit.
- Clearly state if any content comes from a commercial source.

Tell us what you think!



Kindly submit the articles to our editorial team: [Sonia/Vinod](#)

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